7. FREQUENCY RESPONSE OF SERIES AND PARALLEL RESONANCE CIRCUITS

AIM:-

To study the behaviour of series and parallel RLC circuit and determine the resonant frequency band width and Q of the given RLC circuit using multisim simulator.

Apparatus Required

S.NO	components	Specification	Quatity
1.	CRO	10MHz	1
2.	DIB	---	1
3.	Capacitor	0.1μF	1
4.	Resistor	330Ω	1
5.	Brand board	---	---
6.	connecting wire PC with Multisim	---	--
7.	Simulator	---	--

Theory :-

A resonant circuit, also called a tuned circuit consists of L and C. Resonant circuits allow us to select a desired signal from the vast number of signals from the vast number of signals that are around us at any time. A network is in resonance, when the voltage and current are in phase and the input impedance of the network is purely resistive. Considering the parallel RLC circuit, the steady-state admittance offered by the circuit.

$$Y = 1/R + j(\omega C - 1/\omega L)$$

Resonance occurs when the voltage and current at the terminals are in phase. This corresponds to a purely real admittance so that the necessary condition is given by $\omega C - (1/\omega L) = 0$

The resonant condition may be achieved by adjusting L, C or W. Keeping L and C constant the resonant frequency ω_0 is given by:

$$\omega_0 = 1/\sqrt{LC}$$

$$f_0 = 1/2\pi\sqrt{LC}$$

Frequency Response a plot of output voltage or current of a resonance circuit as function of frequency, The ~~response~~. The response

The response reaches a maximum value in the vicinity of the natural resonant frequency, and then drops again to zero as F becomes Infinite. The frequency response is shown in figure the two additional frequencies F_1 and F_2 are also indicated which are called half power frequency locate those Points on the curve at which the voltage response is $1/\sqrt{2}$ or 0.707 times the maximum value. They are used to measure the bond with of the response curve. Thus is called the bond with of the response curve. This is called. Thus is called the half-Power band width of the resonant circuit and is defind as

$$BW = F_2 - F_1$$

Procedure:-

1. Constant circuit shown in figure 1 on bread board.

2. Connect CRD Ch_1 to input and ch_2 to output.

3. Set the input and Ch_2 to output

4. Vary the frequency from 500 Hz to 3KHz in small steps to get a maximum output voltage magnitude The frequency from 500Hz to 3K

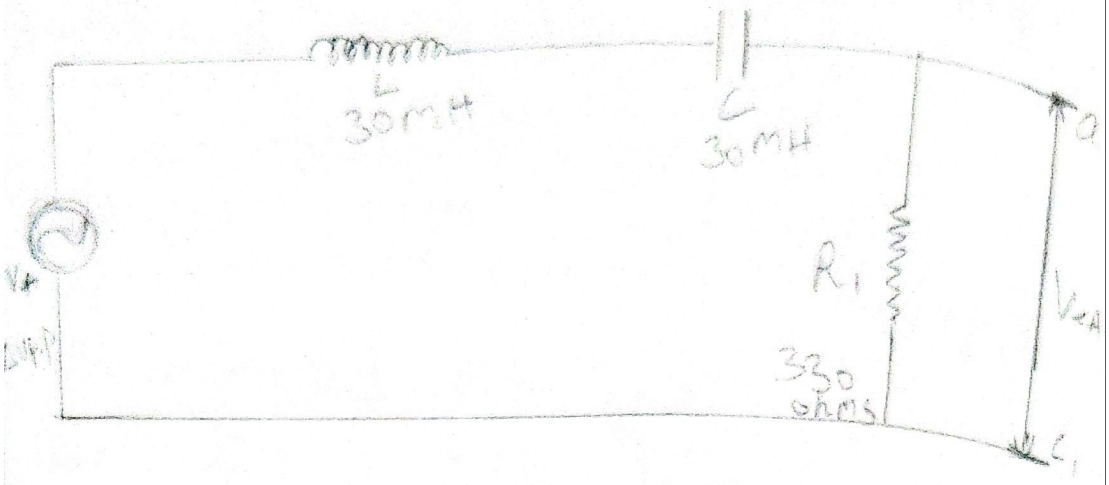
5. If $V_{cut\ off} = V_m/\sqrt{2}$, vary the frequency again untill the output voltage equals $V_{cut\ off}$. this frequency. There should be 2 cut off frequency is the

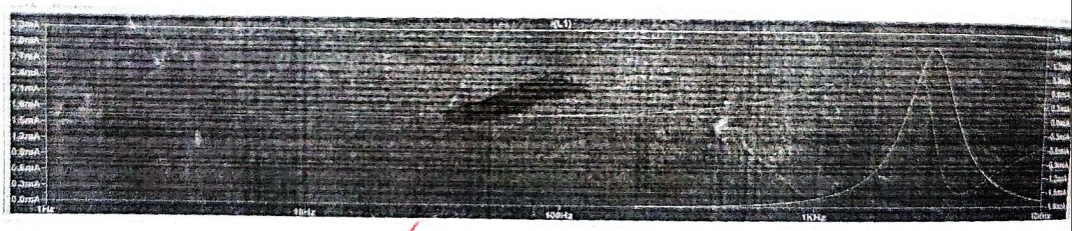
6. Calculate the bandwith by subtracting the 2 cut-off frequency.

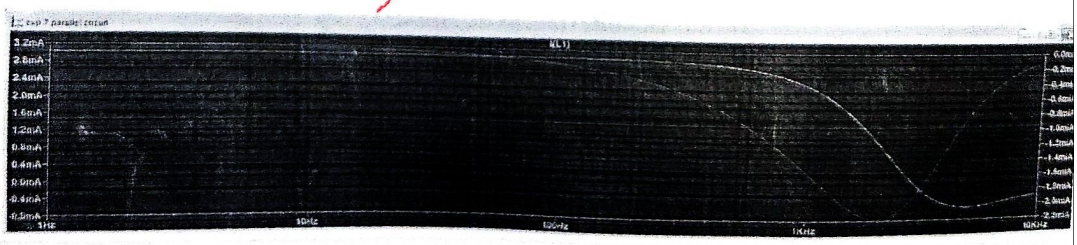
7. Calculated the Q factor

8. Repeated Steps 1 through 7 for circuit ~~circuit~~ shown in figure 2.

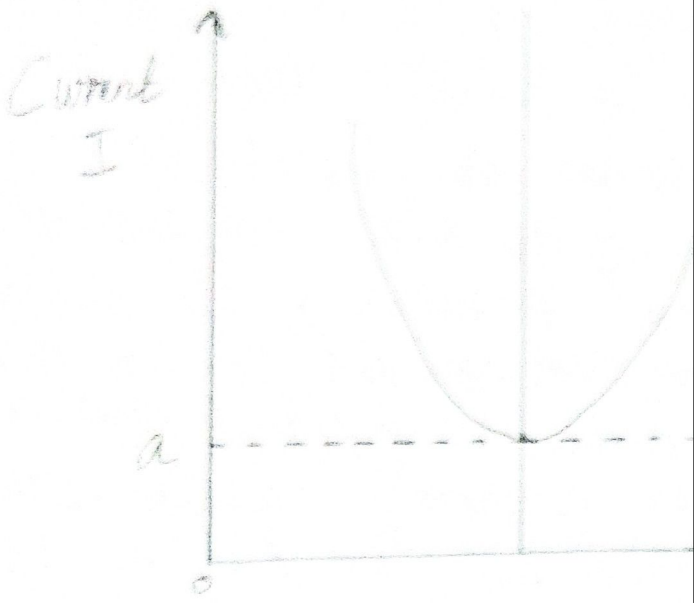
CIRCUIT DIAGRAM:-



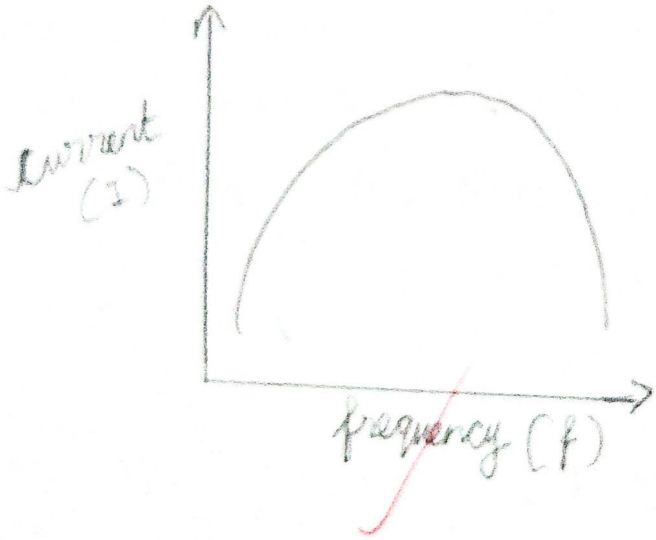




PARALLEL RESONANCES



SERIES RESONANCE



TABULATION : Vin = 4VP-P

Series Resonance		Parallel Resonance	
Frequency (HL)	Output current (MA)	Frequency (HL)	Output current (MA)
500 HL	0.04 MA	500 Hz	2.9 MA
1 KHZ	0.16 MA	1 KHZ	2.54 MA
1.5 KHZ	0.46 MA	1.5 KHZ	1.96 MA
2 KHZ	1.15 MA	2 K HZ	1.26 MA
2.9 KHZ	3 MA	2.9 KHZ	0.002 MA
3.5 KHZ	2.16 MA	3.5 KHZ	0.6 MA.

Calculation :-

Theoretical Calculation

For. Series

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$= \frac{1}{2\times 3.14\sqrt{30\times 10\times 0.1\times 10^{-6}}}$$

$$f_r = 2.9\,KHz$$

$$\text{Bandwith} = \frac{R}{2\pi L}$$

$$= \frac{360}{2\pi\times 30\times 10^{-3}}$$

$$BW = 1.75\,KHz$$

Quantity factor :-

$$Q = \frac{f_r}{BW}$$

$$= \frac{2906}{1750}$$

$$Q = 1.66$$

Parallel circuit

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

Band with,

$$BW = \frac{1}{2\pi RC}$$

$$= \frac{1}{2\times 3.14\times 330\times 0.1\times 10^{-6}}$$

$$BW = 4.82\,KHz$$

Quantity Factor

$$Q = \frac{f_r}{BW}$$

$$Q = \frac{2906}{4820} = 0.6$$

Mark split up :-

S.I.No	Description	marks allotted	marks obtained
1.	Experimental setup / connections	20 marks	18
2.	Observation / Reading	20 marks	18
3.	Calculation / Result	20 marks	18
4.	VIVA	20 marks	18
5.	Record	20 marks	18
		Total 100 marks	90

Result :-

Thus, the phenomenon of resonance in RLC circuit was studied and the following were determined using Multisim simulator

PS